

9.10 Determining How Dilations Affect Area

Lesson Introduction

 Benchmark	MA.912.GR.4.3 Extend previous understanding of scale drawings and scale factors to determine how dilations affect the area of two-dimensional figures and the surface area or volume of three-dimensional figures.		 Learning Target	I can discover the relationship between area and the scale factor of a dilation.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the relationship between area and the scale factor of a dilation? - How does the scale factor of a figure affect the area?			
 Lesson Narrative	The lesson gives students the opportunity to discover how the area of a preimage and an image relate to dilations with given scale factors. 9.10.2 Exploration guides students to make conjectures to discover the relationship between the area and the scale factor of the dilation. After 9.10.2 Exploration, students work through two components to gain practice to master this benchmark.			
 Component Summary	9.10.1 Warm-Up (~5 min.)	Speculating about a directional sign (<i>SE p. 105</i>)	9.10.3 Guided Practice (10-15 min.)	Solving problems using scale factor and area (<i>SE p. 109</i>)
	9.10.2 Exploration (10-15 min.)	Discovering the relationship between the scale factor of a dilation and area (<i>SE p. 105</i>)	9.10.4 Your Turn! (5-10 min.)	Finding the area of a dilated circle (<i>SE p. 110</i>)
	9.10.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 111</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Colored pencils/Pens (Optional) Graphing Utility	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may mistakenly think that the scale factor is the proportion that the area will increase or decrease by. - Students may incorrectly identify a scale factor of less than 1 as an enlargement. - On 9.10.4 Your Turn! question 1, students may incorrectly find the radius of the image.	- Consider allowing students to use a calculator to compute the coordinates that have fractions. - Consider showing applets in GeoGebra or another program that provide students with a visual approach.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns Provide students an opportunity to work 9.10.2 Exploration with either a partner or a small group. The questions and prompts lend themselves for students to take turns working through the guided discovery. Create a structure where students explain their work and their conjectures.	MA.K12.MTR.1.1: Participate in Effortful Learning MA.K12.MTR.1.1 states that students are expected to help and support one another when completing mathematical tasks and to analyze and make sense of the questions. 9.10.2 Exploration gives students the opportunity to practice and apply the skills of active participation.
 Differentiate Instruction	Consider using embedded visuals into 9.10.2 Exploration to help guide students through the component. GeoGebra has a plethora of applets that model scale factor and area to help students better visualize how the scale factor of a dilation affects the area of a two-dimensional figure. Within GeoGebra, search “scale factor and area” and choose an applet that best fits student needs.	
Lesson Notes:		

9.10.1 Warm-Up: Where Is Here?

Component Overview: Students complete three questions involving a picture.



9.10.1 Warm-Up: Where is Here?

A picture of a directional sign is shown.



Consider also asking students, “What do you wonder about the sign?” Allow them the opportunity to share out their answers to gather group discussion and thoughts around this picture, since there is not one correct answer for these questions.

1. What do you notice about the sign?
Answer will vary.
The sign says, “Here”. It gives no true direction.
2. Where do you think the picture was taken?
Answer will vary.
In a city near a river, maybe a European city.
3. The Earth’s average polar circumference is 24,859 miles. What do you think the sign means?
Answer will vary.
It may mean that if someone walks the complete circumference of the Earth’s surface, they will pass this sign.

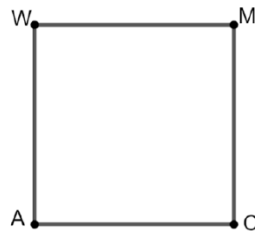
9.10.2 Exploration: Scale Factor and Area

Component overview: 9.10.2 Exploration begins with a one-unit square where students are required to find the area, perform a dilation to discover a pattern, and write a conjecture relating the area of a figure to the scale factor. Then students refine their conjectures by applying it to larger squares, rectangles, and parallelograms. Students are expected to apply previous knowledge of dilations.



9.10.2 Exploration: Scale Factor and Area

1. Square $AWMC$ is shown, where $WM = 1$ unit.



- a. Determine the area of $AWMC$.

$$A = 1 \text{ unit}^2$$

- b. $AWMC$ is dilated using point A as the center of dilation using each of the given scale factors. Complete the table.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation	$\frac{1}{2} \text{ unit}$	$\frac{3}{4} \text{ unit}$	2 units	3 units	4 units
Area of Dilated Square	$\frac{1}{4} \text{ unit}^2$	$\frac{9}{16} \text{ unit}^2$	4 unit ²	9 unit ²	16 unit ²

- c. Write a conjecture about the relationship between the length of the scale factor and the value of the area.

The value of the area of $AWMC$ is equal to the square of the measure of the scale factor.

- d. Determine the area of $AWMC$ if $WM = 4$ units.

$$A = 16 \text{ unit}^2$$



Students should be collaborating throughout the component. Monitor and listen for common student misconceptions to identify which students surface those misconceptions and which students may need help with clarifications.



Consider asking students the following questions as they are working on the table in question 1b:

- Why is the side length of the dilation the same as the scale factor? (The figure is a square, and the preimage is 1.)
- How do you find the area of the dilated square?

9.10.2 Exploration: Scale Factor and Area (continued)

2. Suppose square $DTSC$ is dilated using point D as the center of dilation for each of the given scale factors. Complete the table for $DTSC$ given that $TS = 4$ units.

Scale Factor	$\frac{1}{2}$	$\frac{3}{4}$	2	3	4
Side Length of Dilation	2 units	3 units	8 units	12 units	16 units
Area of Dilated Square	4 units ²	9 units ²	64 units ²	144 units ²	256 units ²

- a. Review your conjecture from the previous problem and make any revisions, if needed.
The value of the area of $DTSC$ is equal to (side length $\times$ scale factor)².
- b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

Summary of Partner's Conjecture	Partner's Initials
<i>Answers will vary.</i>	

- c. Review your conjecture and make any revisions, if needed.
Answers will vary.



For a kinesthetic approach to 9.10.2 Exploration, consider dividing the classroom into three groups, one for the square, the rectangle, and the parallelogram. Have students become an expert on their figure. Regroup the classroom into groups of three where each student shares their results of their questions.

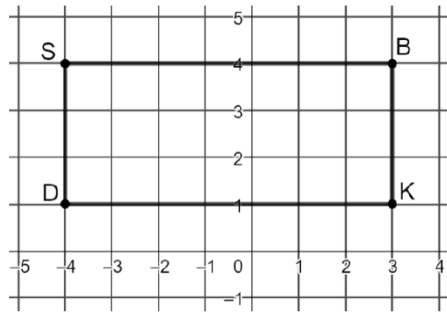


Ask the following questions to prompt student thinking:

- Does the conjecture need to be revised? Why or why not?

9.10.2 Exploration: Scale Factor and Area (continued)

3. Rectangle $SBKD$ is shown.



- a. Using the coordinate grid, find the area of the rectangle.

Area= 21 square units

- b. Dilate $SBKD$ by a scale factor of 3 centered at the origin. Write the new coordinates in the table.

Point	Preimage	Image
S	$(-4, 4)$	$(-12, 12)$
B	$(3, 4)$	$(9, 12)$
K	$(3, 1)$	$(9, 3)$
D	$(-4, 1)$	$(-12, 3)$



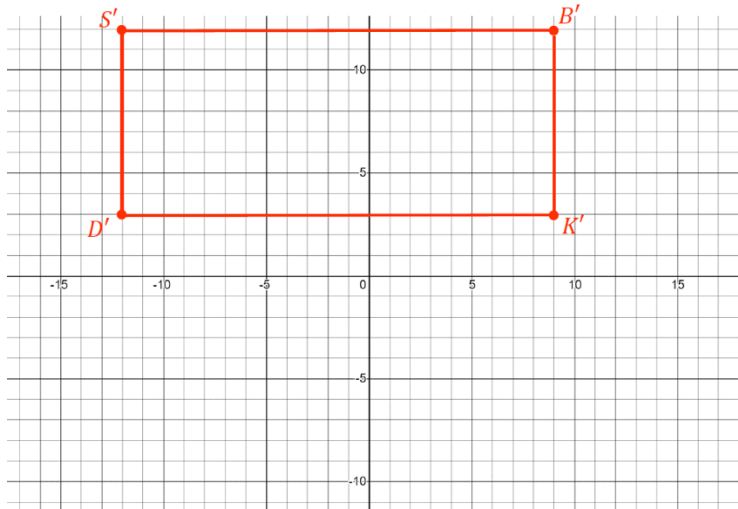
Does the conjecture need to be revised? Why or why not?



What does the dilation do to the figure? How is this different than a translation?

9.10.2 Exploration: Scale Factor and Area (continued)

4. Graph $S'B'K'D'$ on the coordinate plane.



- a. The area of rectangle $S'B'K'D'$ is 189 square units.
- b. Does the conjecture you wrote about the relationship between area and scale factor when dilating a square hold true for rectangle $SBKD$?

Answer will vary.

No, for a rectangle, the area of the dilated rectangle equals the product of the area of the pre-image and the square of the measure of the scale factor.

Figure	Length	Width	Area
Pre-Image	3 $\rightarrow$.3	7 $\rightarrow$.3	21 units ²
Image	9 $\leftarrow$	21 $\leftarrow$	189 units ² $\leftarrow$.9



Allow students to use colored pencils and/or pens to graph on the coordinate plane to make the graphs more visual.

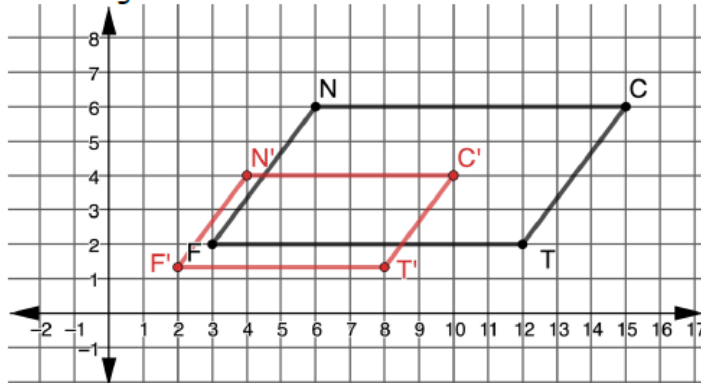


Consider asking the following question to promote student discussion:

- Why does the conjecture about the relationship between area and scale factor for the square not hold true for the rectangle?*

9.10.2 Exploration: Scale Factor and Area (continued)

5. Parallelogram $FNCT$ is shown.



Before having students fill in the table to find the coordinates of the image, have students make a prediction about what they think the area of the image will be when compared to the preimage.

- a. What is the area of $FNCT$, in square units?
The area of $FNCT$ is 36 units².
- b. Dilate $FNCT$ by a scale factor of $\frac{2}{3}$ centered at the origin. Write the coordinates of the preimage and image in the table.

Point	Preimage	Image
F	$(3,2)$	$(2, \frac{4}{3})$
N	$(6,6)$	$(4,4)$
C	$(15,6)$	$(10,4)$
T	$(12,2)$	$(8, \frac{4}{3})$

- c. Plot the coordinates of $F'N'C'T'$ on the same coordinate plane as $FNCT$.
- d. What is the relationship between the area of $FNCT$ and $F'N'C'T'$?
The area of $F'N'C'T'$ is 16 unit², which is $\frac{4}{9}$ the area of $FNCT$. The value $\frac{4}{9}$ is the square of the scale factor, $\frac{2}{3}$.

9.10.2 Exploration: Scale Factor and Area (continued)

- e. How can the scale factor of $\frac{2}{3}$ be used to determine the area of $F'N'C'T'$ from the area of $FNCT$?

Since each dimension is $\frac{2}{3}$ of the original figure, the area of parallelogram $F'N'C'T'$ will be $(\frac{2}{3} \times \text{base})(\frac{2}{3} \times \text{height})$, which means that the area of $F'N'C'T' = (\frac{2}{3})(\frac{2}{3})(\text{area of } FNCT) = (\frac{4}{9})(\text{area of } FNCT)$.



- Does the conjecture for the rectangle hold true for the parallelogram? (Yes.)
- How do you calculate the ratio of the areas of parallelogram $FNCT$ and parallelogram $F'N'C'T'$? (Divide the area of the preimage by the area of the image.)

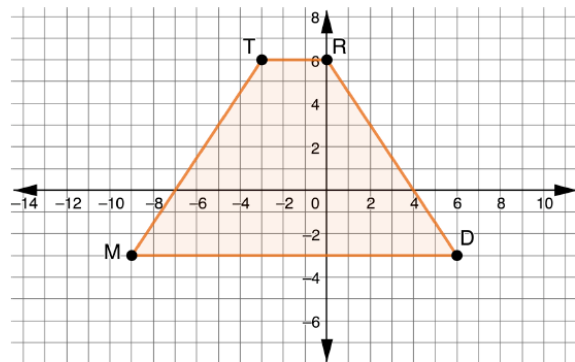
9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area

Component Overview: After 9.10.2 Exploration, students will apply the skills and conjectures to find the area of a trapezoid preimage and then perform dilations to find the area and determine the effects that the scale factor has on the area.



9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area

1. Trapezoid $MTRD$ is shown on the coordinate plane.



- a. Find the area of $MTRD$.

$$A = \frac{1}{2}(15 + 3)(9)$$

The area of $MTRD$ is 81 square units.

- b. Find the coordinates of $MTRD$ given the scale factors in the table. Use the preimage coordinates to find the coordinates of the images. Both dilations are centered at $(0,0)$.

Point	Preimage	Dilation $k = \frac{1}{6}$ ($M'T'R'D'$)	Dilation $k = \frac{4}{3}$ ($M''T''R''D''$)
M	$(-9, -3)$	$(-\frac{3}{2}, -\frac{1}{2})$	$(-12, -4)$
T	$(-3, 6)$	$(-\frac{1}{2}, 1)$	$(-4, 8)$
R	$(0, 6)$	$(0, 1)$	$(0, 8)$
D	$(6, -3)$	$(1, -\frac{1}{2})$	$(8, -4)$



What is something you notice about this trapezoid? What is something that you wonder?



Allow students who need extra support with fractions to use a calculator.

9.10.3 Guided Practice: Solving Problems Involving Scale Factor and Area (continued)

- c. Find the area of $M'T'R'D'$.

$$A = \frac{1}{2}(2.5 + 0.5)(1.5)$$

The area of trapezoid $M'T'R'D'$ is 2.25 unit^2 .

- d. What effects does the scale factor of $\frac{1}{6}$ have on the original area?

The product of the square factor and the area of the pre-image is equal to the area of the image.

$$\left(\frac{1}{6}\right)^2 \cdot 81 = 2.25 \text{ unit}^2$$

- e. Find the area of $M''T''R''D''$.

$$A = \frac{1}{2}(20 + 4)(12)$$

The area of $M''T''R''D''$ is 144 unit^2 .

- f. What effects does the scale factor of $\frac{4}{3}$ have on the original area?

The product of the square factor and the area of the pre-image is equal to the area of the image.

$$\left(\frac{4}{3}\right)^2 \cdot 81 = 144 \text{ unit}^2$$

- g. How does the area of the trapezoid with a scale factor of $\frac{1}{6}$ differ from the area of the trapezoid with a scale factor of $\frac{4}{3}$?

Answers will vary. The area of a trapezoid with a scale factor of $\frac{1}{6}$ will be much smaller than the area of the original trapezoid. The area of a trapezoid with a scale factor of $\frac{4}{3}$ will be larger than the area of the original trapezoid.



Allow students to discuss questions 1d, 1f, and 1g with a partner or whole group. Consider having students use the Literacy and Language Routine Take Turns when sharing their responses to the questions.

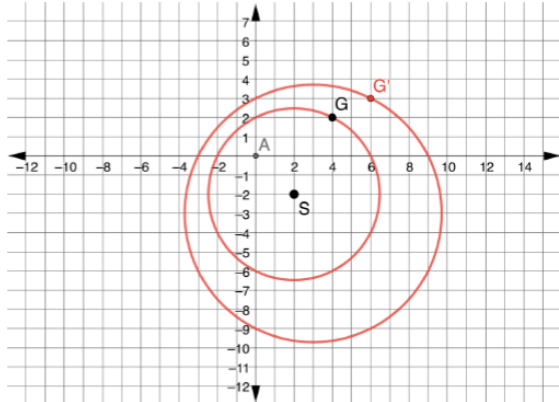
9.10.4 Your Turn!

Component Overview: Students are given a circle with a radius that is a square root. Students graph the image and find the areas of the preimage and the image.



9.10.4 Your Turn!

1. Circle S is shown with a radius of $\sqrt{20}$.



- a. Dilate circle S using a scale factor of 1.5 centered at the origin, and graph the resulting image on the coordinate plane.

See the graph.

- b. Find the area of the circles.

Area of Preimage	Area of Image
$r = \sqrt{20}$	$r' = \sqrt{45}$
$A = \pi r^2$	$A = \pi (r')^2$
$A = \pi (\sqrt{20})^2$	$A = \pi (\sqrt{45})^2$
$A = 20\pi$	$A = 45\pi$



When students are finding the radius of the image, students may want to multiply 20 by 1.5, which would give students 30, but this will yield an incorrect radius. Students must multiply $(\sqrt{20})(1.5)$, which equals $\sqrt{45}$.



Remind students of the formula of the area of the circle if necessary.



For students who finish early, challenge them to create a third circle with another dilation. Have students then calculate the area and the radius of the circle and compare.

9.10.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



9.10.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about how dilations affect area.

Answers will vary.